

Chapter 5: Mathematics Logic

Chapter-End Activities

Activity 1: Math Puzzle Challenge

Objective

To strengthen students mathematical reasoning, observation, and problem-solving skills through fun number and shape puzzles.

Materials Required

- Math Puzzle Worksheets
- Number Cards
- Shape Cards
- Whiteboard & Marker
- Timer
- Scoreboard

Instructions

1. Divide the class into **4–5 teams**.
2. Distribute different math puzzles to each team.
3. Allow **5–7 minutes** to solve the puzzles.
4. Teams present their answers and explain their reasoning.
5. Award points for correct answers and logical explanations.

Puzzle Activities

Number Pattern

5, 10, 15, __, 25

Answer: 20

Shape Pattern

☐ ☒ ☐ ☒ ☐ __

Answer: ■

Missing Number

12, 24, __, 48

Answer: 36

Even or Odd

Identify whether the following numbers are **Even** or **Odd**:

- 48
 - 35
 - 92
 - 67
 - 100
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Ascending Order

Arrange the numbers:

42, 18, 56, 29

Answer: 18, 29, 42, 56

Assessment Criteria

- Accuracy
 - Logical Thinking
 - Observation Skills
 - Problem-Solving Ability
 - Team Participation
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Activity 2: Logic Quiz

Objective

To assess students' understanding of mathematical concepts, logical reasoning, number patterns, and mental mathematics.

Materials Required

- Quiz Cards
- Whiteboard
- Timer
- Scoreboard

Instructions

1. Conduct the quiz individually or in teams.
2. Ask one question at a time.
3. Allow **30 seconds** to answer each question.
4. Award **1 point** for each correct answer.
5. Announce the winners after all rounds.

Sample Quiz Questions

1. What comes next: **20, 30, 40, __?**
2. Which number is greater: **95 or 59?**
3. What is **8×9** ?
4. What is **$81 \div 9$** ?
5. Name two odd numbers.
6. What is a number pattern?
7. What is skip counting?
8. Which direction is opposite to **East**?
9. Complete the analogy: **Bird : Nest :: Bee : _____**
10. Name one strategy game you learned.

Assessment Criteria

- Mathematical Knowledge
 - Speed
 - Accuracy
 - Logical Reasoning
 - Confidence
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Activity 3: Team Problem-Solving Competition

Objective

To develop teamwork, communication, and problem-solving skills through real-life mathematical challenges.

Materials Required

- Problem-Solving Worksheets
- Chart Paper
- Marker Pens
- Whiteboard
- Timer

Instructions

1. Divide students into teams of **4–5 members**.
2. Give each team a worksheet containing mathematical word problems.
3. Allow **8–10 minutes** to discuss and solve the problems.
4. One representative from each team presents the solutions.
5. Encourage students to explain the steps used to arrive at the answers.

Problem-Solving Activities

Problem 1

A school library has **250 books**. The school buys **75 more books**.

How many books are there now?

Answer: 325 books

Problem 2

A farmer has **120 mangoes**. He sells **45 mangoes**.

How many mangoes are left?

Answer: 75 mangoes

Problem 3

A bus can carry **60 passengers**. **48 passengers** are already seated.

How many seats are still available?

Answer: 12 seats

Problem 4

A student reads **18 pages** every day.

How many pages will the student read in **5 days**?

Answer: 90 pages

Problem 5

A shopkeeper has **150 pencils**. He sells **65 pencils**.

How many pencils remain?

Answer: 85 pencils

Assessment Criteria

- Problem Analysis
 - Correct Mathematical Operations
 - Teamwork
 - Communication Skills
 - Presentation Skills
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Oral Assessment

Objective

To evaluate students' mathematical reasoning, mental calculation, and communication skills.

Sample Oral Questions

- What is a number pattern?
- Name five even numbers.
- What is skip counting?
- Which number comes after **499**?
- What is **15×6** ?
- What is **$96 \div 8$** ?
- What is estimation?
- Why is logical reasoning important?
- Give one example of mathematics used in daily life.
- Name one strategy game you enjoyed.

Assessment Criteria

- Fluency
 - Accuracy
 - Confidence
 - Logical Thinking
 - Mathematical Vocabulary
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Practical Evaluation

Objective

To assess students' practical understanding of all concepts learned in **Chapter 5: Mathematics Logic**.

Practical Tasks

Students should be able to:

- Complete number patterns and sequences.
- Identify even and odd numbers.
- Perform mental addition, subtraction, multiplication, and division.
- Solve visual puzzles and brain teasers.

- Complete a basic Sudoku puzzle.
- Compare and arrange numbers correctly.
- Solve real-life mathematical word problems.
- Explain their reasoning confidently.

Practical Evaluation Rubric

Assessment Area	Marks
Number Patterns & Sequences	15
Mental Mathematics	20
Logical Reasoning & Visual Puzzles	20
Problem-Solving Skills	20
Team Participation	10
Oral Reasoning Assessment	10
Confidence & Presentation	5
Total	100 Marks

Awards & Recognition

Students may be recognized for outstanding performance in different areas:

- 🏆 Best Problem Solver Award
 - 🧩 Best Logic Thinker Award
 - ⚡ Fastest Mental Maths Award
 - 🤝 Best Team Player Award
 - 🎯 Best Puzzle Solver Award
 - 🌟 Most Improved Mathematics Learner Award
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Chapter Learning Outcomes

After completing the **Chapter-End Activities** for **Chapter 5: Mathematics Logic**, students will be able to:

- Identify and complete **number patterns and sequences** accurately.
- Apply **logical reasoning** to solve visual puzzles, brain teasers, and mathematical challenges.
- Perform **mental mathematics** with improved speed, accuracy, and confidence.
- Solve real-life mathematical problems using appropriate strategies and logical thinking.

- Demonstrate effective **teamwork, communication, analytical thinking, and decision-making skills** during collaborative activities.
- Develop confidence in applying mathematical concepts independently in classroom activities and everyday life.